

BASIC ELECTRONICS ENGINEERING

MODULE 1

SEMICONDUCTOR DEVICES AND

APPLICATIONS

Module I:

Semi-conductor devices and applications: p-n junction diode - Characteristics and Parameters

Half-wave rectifier, Full-wave rectifiers and filters, Voltage regulators, Output resistance and voltage regulation, Voltage multipliers. (Only concepts and working and principle. No mathematical derivations).

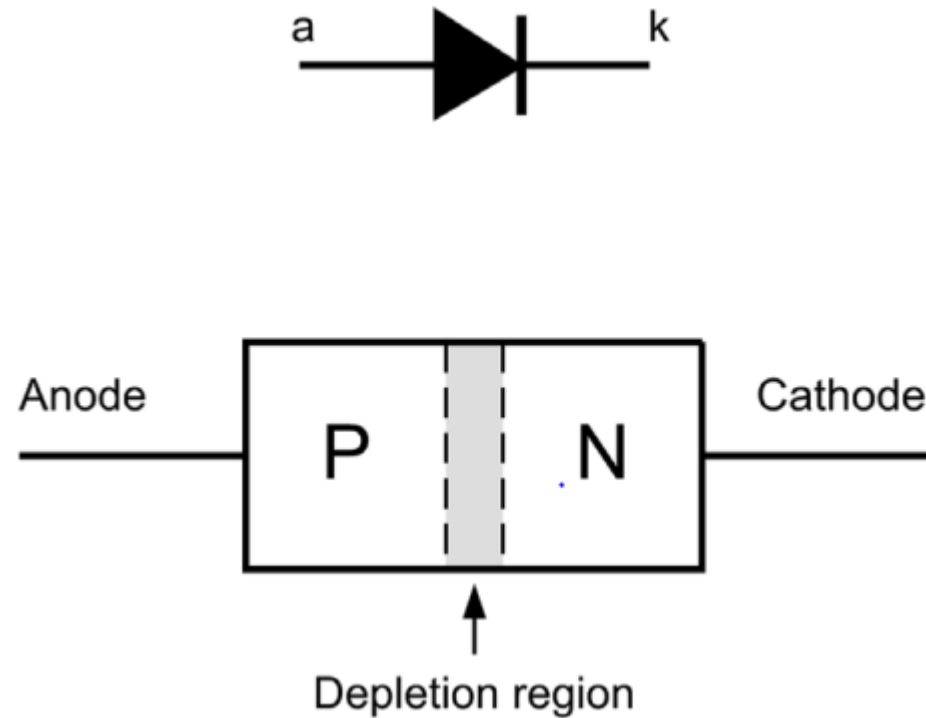
Amplifiers – Types of amplifiers, Gain, Input and output resistance, Frequency response, Bandwidth, Phase shift, Negative feedback, multi-stage amplifiers. (Only elementary ideas. Mathematical treatment is not envisaged)

- The process of introducing (impurity) element into the lattice of a pure material is called **doping**.
- When the pure material has been doped with an impurity with five electrons in its valence shell (i.e. a pentavalent impurity) it will become an N-type material.
- If, however, the pure material is doped with an impurity having three electrons in its valence shell (i.e. a trivalent impurity) it will become a P-type material.
- N-type semiconductor material contains an excess of negative charge carriers, and P-type material contains an excess of positive charge carriers.

- When a junction is formed between N-type and P-type semiconductor materials, the resulting device is called a diode.
- This component offers an extremely low resistance to current flow in one direction and an extremely high resistance to current flow in the other.
- This characteristic allows the diode to be used in applications that require a circuit to behave differently according to the direction of current flowing in it.
- An ideal diode would pass an infinite current in one direction and no current at all in the other direction. In addition, the diode would start to conduct current when the smallest of voltages was present.

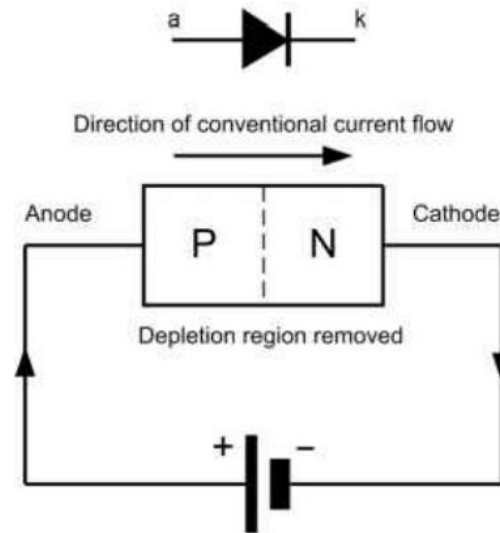
- If the P-type semiconductor material is made positive relative to the N-type material by an amount greater than its forward threshold voltage (about 0.6 V if the material is silicon and 0.2 V if the material is germanium), the diode will freely pass current. If, on the other hand, the P-type material is made negative relative to the N-type material, virtually no current will flow unless the applied voltage exceeds the maximum (breakdown) voltage that the device can withstand. Note that a normal diode will be destroyed if its reverse breakdown voltage is exceeded

- A semiconductor junction diode is shown below.



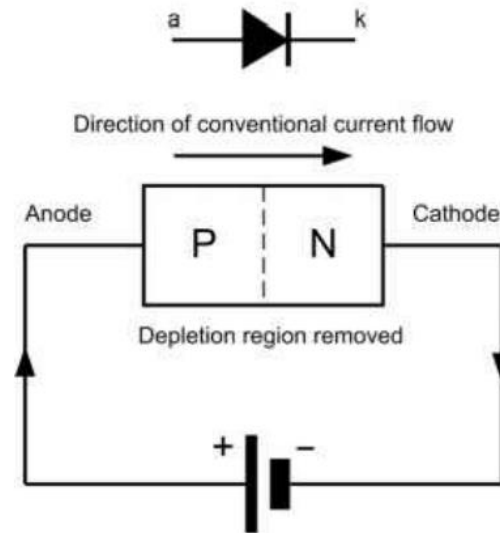
- The connection to the P-type material is referred to as the anode while that to the N-type material is called the cathode.

- With no externally applied potential, electrons from the N-type material will cross into the P-type region and fill some of the vacant holes. This action will result in the production of a region either side of the junction in which there are no free charge carriers. This zone is known as the **depletion region**.



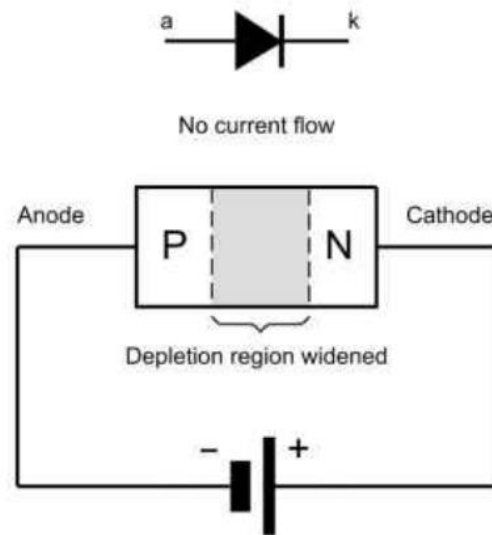
(a) Forward biased P-N junction

- Fig. below shows a junction diode in which the anode is made positive with respect to the cathode. In this forward-biased condition, the diode freely passes current.



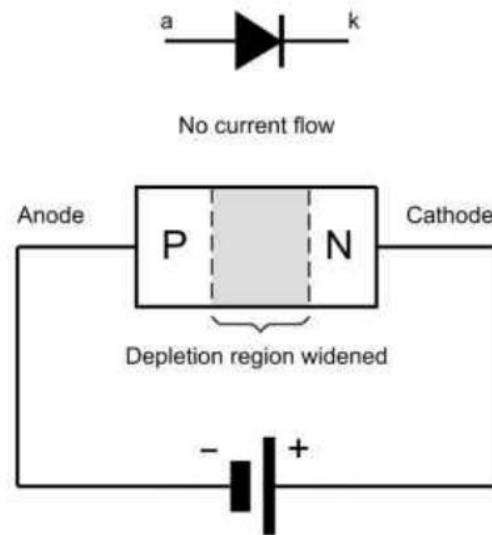
(a) Forward biased P-N junction

- Figure below shows reverse biased P-N junction. In this reverse-biased condition, the diode passes a negligible amount of current. In the freely conducting forward-biased state, the diode acts rather like a closed switch. In the reverse-biased state, the diode acts like an open switch.



(b) Reverse biased P-N junction

- If a positive voltage is applied to the P-type material, the free positive charge carriers will be repelled and they will move away from the positive potential towards the junction and the negative potential applied to the N-type material will cause the free negative charge carriers to move away from the negative potential towards the junction.



(b) Reverse biased P-N junction

- When the positive and negative charge carriers arrive at the junction, they will attract one another and combine (recall that unlike charges attract).
- As each negative and positive charge carrier combine at the junction, a new negative and positive charge carrier will be introduced to the semiconductor material from the voltage source. As these new charge carriers enter the semiconductor material, they will move towards the junction and combine. Thus, current flow is established and it will continue for as long as the voltage is applied.

- the forward threshold voltage must be exceeded before the diode will conduct.
- The forward threshold voltage must be high enough to completely remove the depletion layer and force charge carriers to move across the junction. With silicon diodes, this forward threshold voltage is approximately 0.6 V to 0.7 V.
- With germanium diodes, the forward threshold voltage is approximately 0.2 V to 0.3 V.

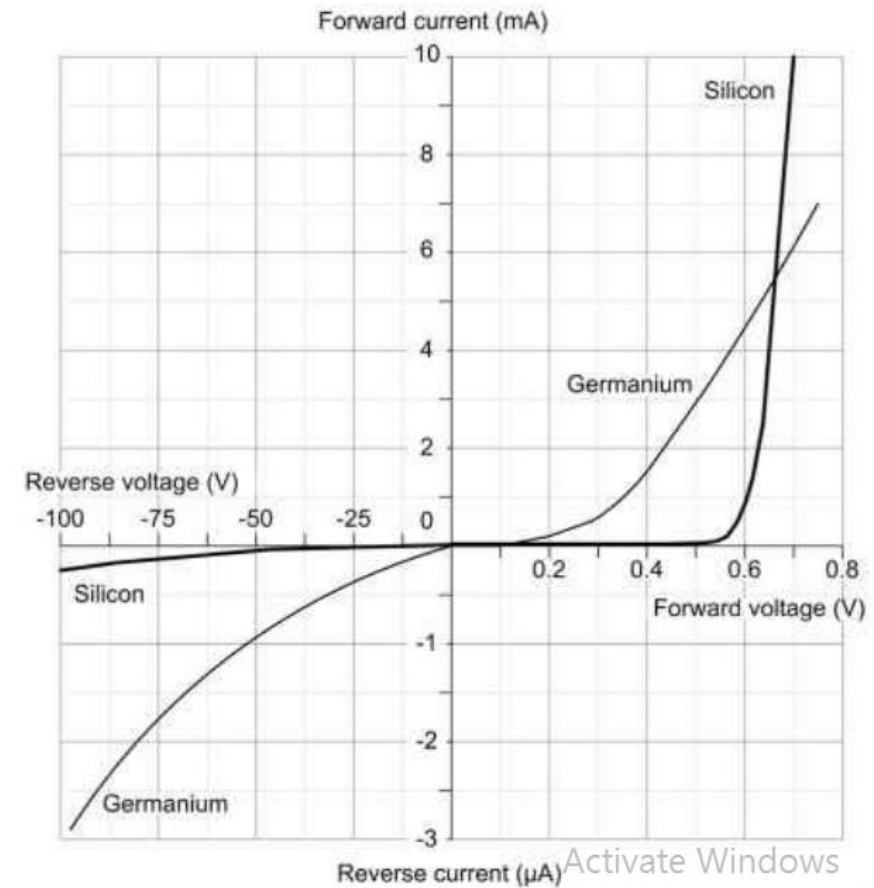
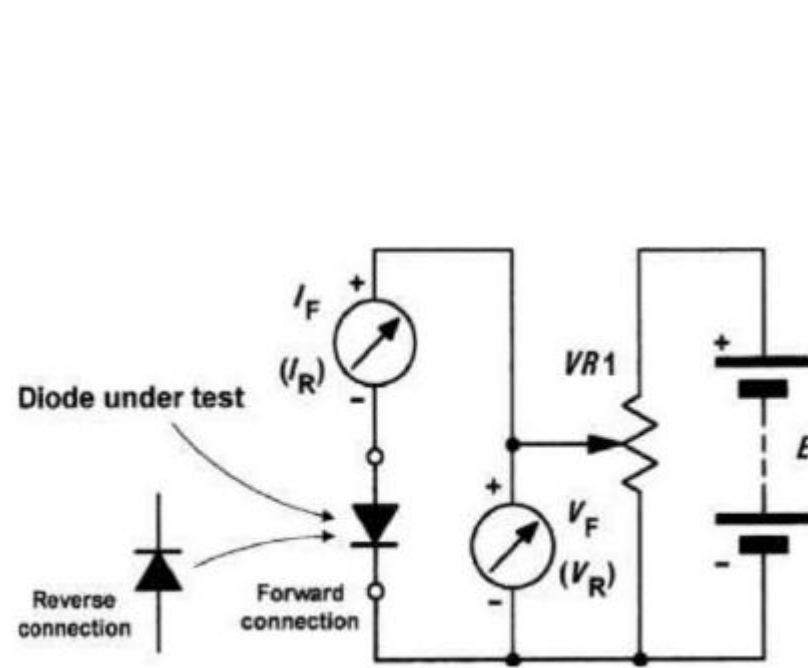


Fig. 1 shows a test circuit for obtaining diode characteristics (note that the diode must be reverse connected in order to obtain the reverse characteristic).

- In the case of a reverse-biased diode, the P-type material is negatively biased relative to the N-type material. In this case, the negative potential applied to the P-type material attracts the positive charge carriers, drawing them away from the junction.
- Likewise, the positive potential applied to the N-type material attracts the negative charge carriers away from the junction. This leaves the junction area depleted; virtually no charge carriers exist. Therefore, the junction area becomes an insulator, and current flow is inhibited.
- The reverse bias potential may be increased to the reverse breakdown voltage for which the particular diode is rated. As in the case of the maximum forward current rating, the reverse breakdown voltage is specified by the manufacturer. The reverse breakdown voltage is usually very much higher than the forward threshold voltage. A typical general-purpose diode may be specified as having a forward threshold voltage of 0.6 V and a reverse breakdown voltage of 200 V. If the latter is exceeded, the diode may suffer irreversible damage. It is also worth noting that, where diodes are designed for use as rectifiers, manufacturers often quote peak inverse voltage (PIV) or maximum reverse repetitive voltage (VRRM) rather than maximum reverse breakdown voltage.

Example

The characteristic shown in Fig. 5.8 refers to a germanium diode. Determine the resistance of the diode when (a) the forward current is 2.5 mA and (b) when the forward voltage is 0.65 V.

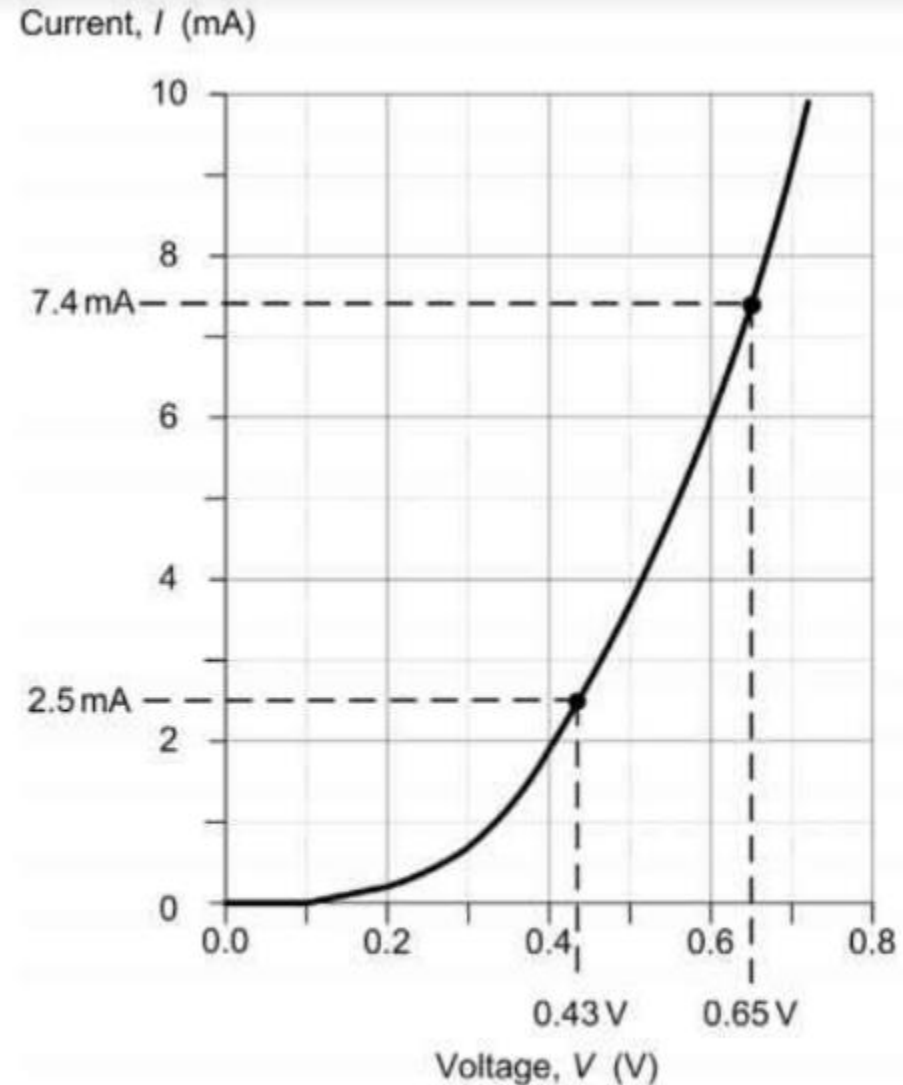
Solution

- (a) When $I_F = 2.5$ mA the corresponding value of V_F can be read from the graph. This shows that $V_F = 0.43$ V. The resistance of the diode at this point on the characteristic will be given by:

$$R = \frac{V_F}{I_F} = \frac{0.43 \text{ V}}{2.5 \text{ mA}} = 172 \, \Omega$$

- (b) When $V_F = 0.65$ V the corresponding value of I_F can be read from the graph. This shows that $I_F = 7.4$ mA. The resistance of the diode at this point on the characteristic will be given by:

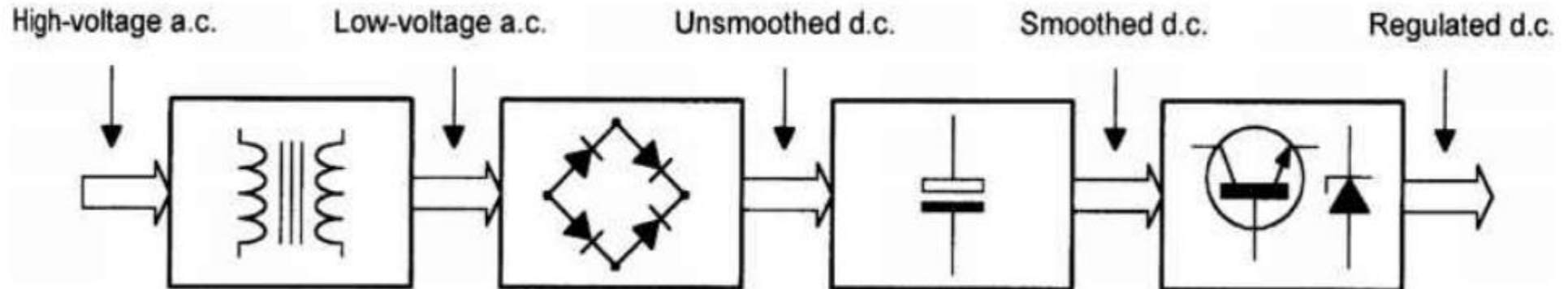
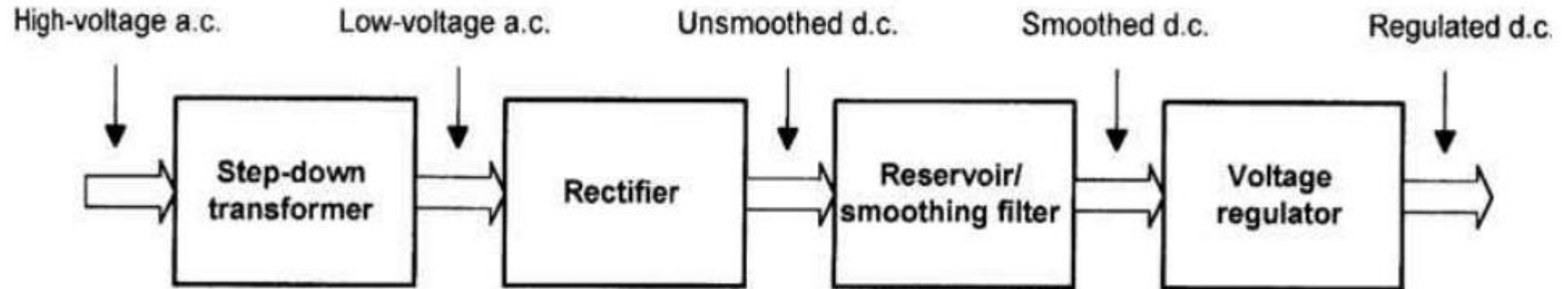
$$R = \frac{V_F}{I_F} = \frac{0.65 \text{ V}}{7.4 \text{ mA}} = 88 \, \Omega$$



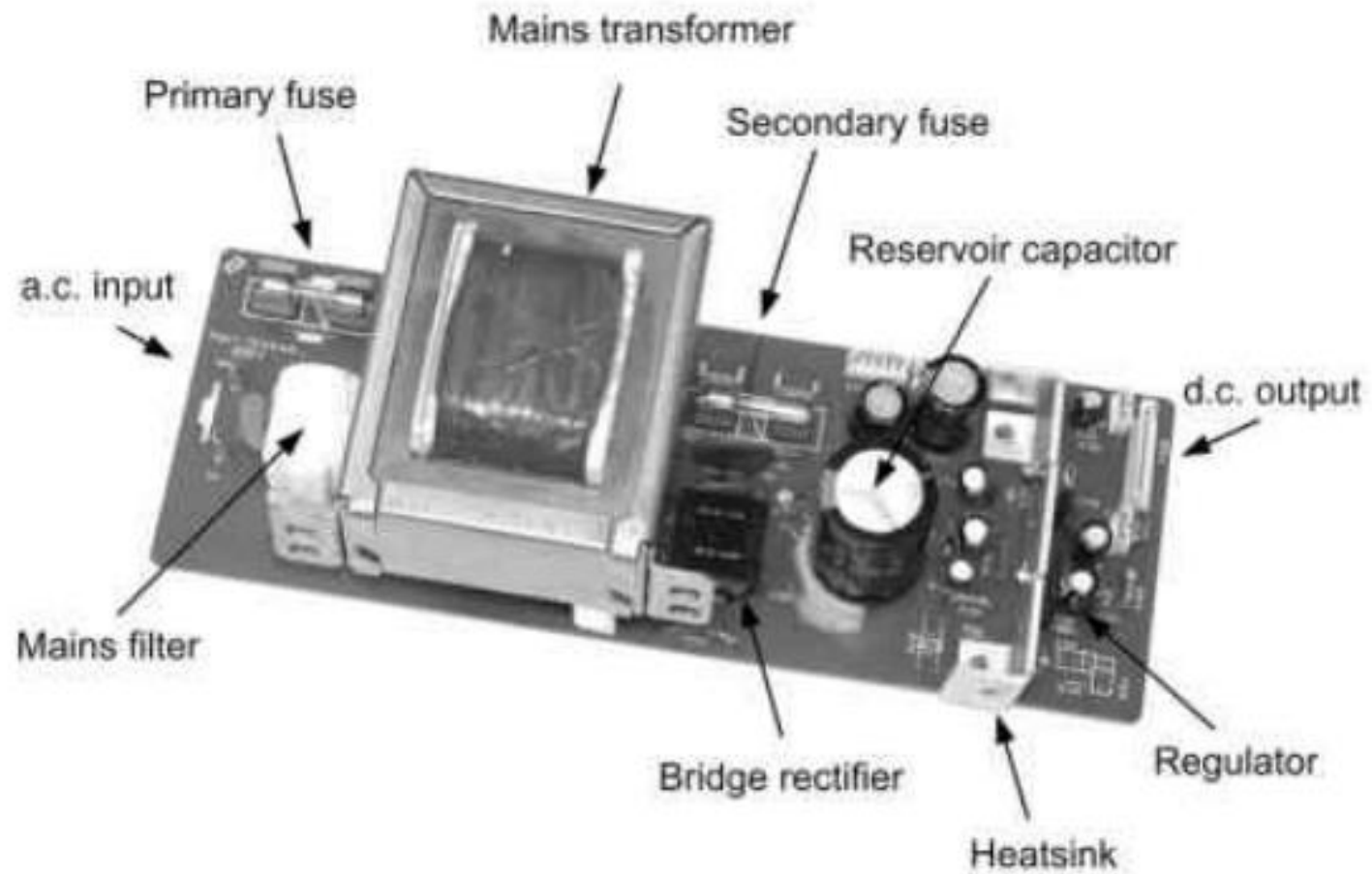
(a) At 0.43 V, $R = 0.43 \text{ V} / 2.5 \text{ mA} = 172 \, \Omega$

(b) At 0.65 V, $R = 0.65 \text{ V} / 7.4 \text{ mA} = 88 \, \Omega$

POWER SUPPLY



POWER SUPPLY

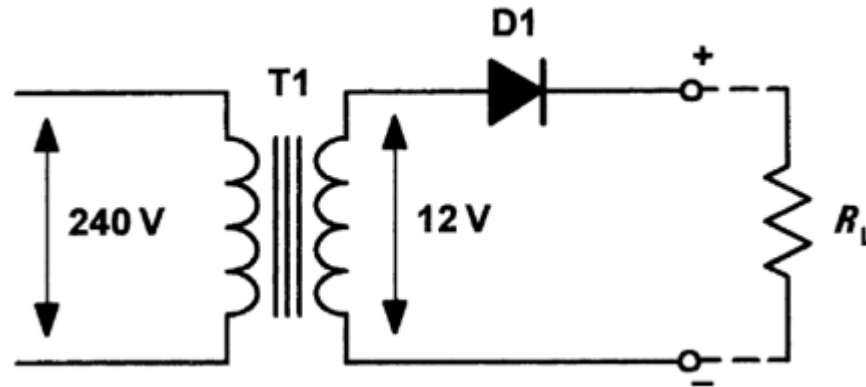


Circuit diagram of a power supply

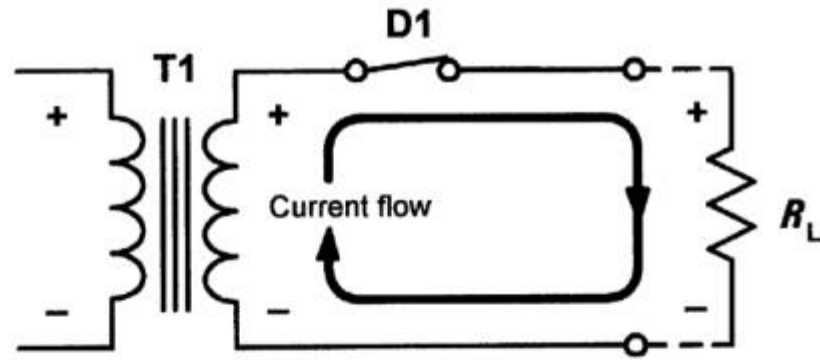
Rectifiers

Semiconductor diodes are commonly used to convert alternating current (a.c.) to direct current (d.c), in which case they are referred to as rectifiers. The simplest form of rectifier circuit makes use of a single diode and, since it operates on only either positive or negative half-cycles of the supply, it is known as a half-wave rectifier.

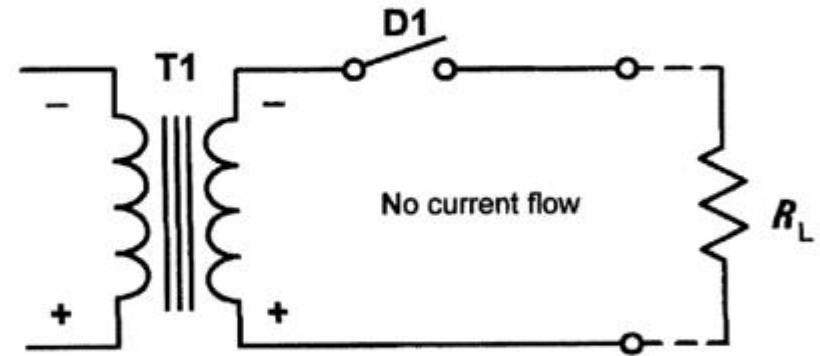
HALF WAVE RECTIFIER



Mains voltage (220 to 240 V) is applied to the primary of a step-down transformer (T1). The secondary of T1 steps down the 240 V r.m.s. to 12 V r.m.s. (the turns ratio of T1 will thus be 240/12 or 20:1). Diode D1 will only allow the current to flow in the direction shown (i.e. from cathode to anode). D1 will be forward biased during each positive half-cycle (relative to common) and will effectively behave like a closed switch. When the circuit current tries to flow in the opposite direction, the voltage bias across the diode will be reversed, causing the diode to act like an open switch. The switching action of D1 results in a pulsating output voltage which is developed across the load resistor (R_L). Since the mains supply is at 50 Hz, the pulses of voltage developed across R_L will also be at 50 Hz even if only half the a.c. cycle is present. During the positive half-cycle, the diode will drop the 0.6 V to 0.7 V forward threshold voltage normally associated with silicon diodes.



(a)



(b)

(a) Half-wave rectifier circuit with D1 conducting (positive-going half-cycles of secondary voltage); (b) half-wave rectifier with D1 not conducting (negative-going half-cycles of secondary voltage)

During the negative half-cycle the peak a.c. voltage will be dropped across D1 when it is reverse biased. This is an important consideration when selecting a diode for a particular application. Assuming that the secondary of T1 provides 12 V r.m.s., the peak voltage output from the transformer's secondary winding is given by:

$$V_{pk} = 1.414 \times V_{r.m.s.} = 1.414 \times 12 \text{ V} = 16.97 \text{ V}$$

The peak voltage applied to D1 will thus be approximately 17 V. The negative half-cycles are blocked by D1 and thus only the positive half cycles appear across RL . Note, however, that the actual peak voltage across RL will be the 17 V positive peak being supplied from the secondary on T1, minus the 0.7 V forward threshold voltage dropped by D1. In other words, positive half-cycle pulses having a peak amplitude of 16.3 V will appear across RL .

Example question

A mains transformer having a turns ratio of 44:1 is connected to a 220 V r.m.s. mains supply. If the secondary output is applied to a half-wave rectifier, determine the peak voltage that will appear across a load.

Solution

The r.m.s. secondary voltage will be given by:

$$V_S = V_P / 44 = 220 / 44 = 5 \text{ V}$$

The peak voltage developed after rectification will be given by:

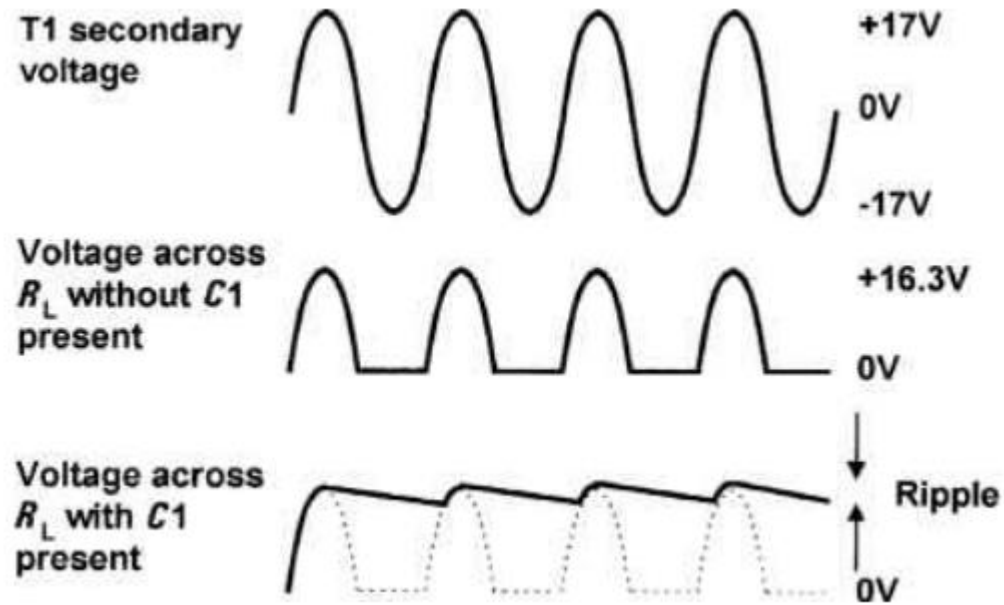
$$V_{PK} = 1.414 \times 5 \text{ V} = 7.07 \text{ V}$$

Assuming that the diode is a silicon device with a forward voltage drop of 0.6 V, the actual peak voltage dropped across the load will be:

$$V_L = 7.07 \text{ V} - 0.6 \text{ V} = 6.47 \text{ V}$$

Reservoir and smoothing circuits

- The capacitor, C_1 , has been added to ensure that the output voltage remains at, or near, the peak voltage even when the diode is not conducting. When the primary voltage is first applied to T1, the first positive half-cycle output from the secondary will charge C_1 to the peak value seen across R_L . Hence C_1 charges to 16.3 V at the peak of the positive half-cycle.
- Because C_1 and R_L are in parallel, the voltage across R_L will be the same as that across C_1 .
- The time required for C_1 to charge to the maximum (peak) level is determined by the charging circuit time constant (the series resistance multiplied by the capacitance value).
- In this circuit, the series resistance comprises the secondary winding resistance together with the forward resistance of the diode and the (minimal) resistance of the wiring and connections. Hence C_1 charges very rapidly as soon as D1 starts to conduct. The time required for C_1 to discharge is, in contrast, very much greater.
- The discharge time constant is determined by the capacitance value and the load resistance, R_L . In practice, R_L is very much larger than the resistance of the secondary circuit and hence C_1 takes an appreciable time to discharge. During this time, D1 will be reverse biased and will thus be held in its non-conducting state. As a consequence, the only discharge path for C_1 is through R_L . C_1 is referred to as a reservoir capacitor. It stores charge during the positive half cycles of secondary voltage and releases it during the negative half-cycles. This circuit is thus able to maintain a reasonably constant output voltage across R_L . Even so, C_1 will discharge by a small amount during the negative half-cycle periods from the transformer secondary.

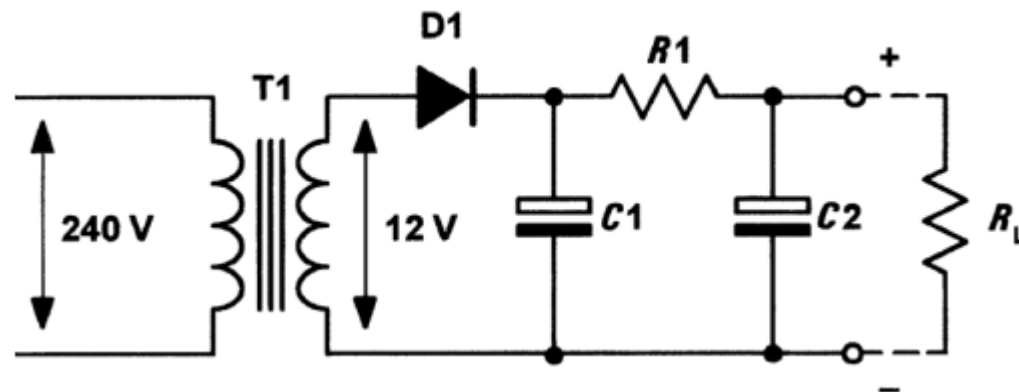


A simple half-wave rectifier circuit with reservoir capacitor

- Figure above shows the secondary voltage waveform together with the voltage developed across R_L with and without C_1 present. This gives rise to a small variation in the d.c. output voltage (known as ripple). Since ripple is undesirable we must take additional precautions to reduce it. One obvious method of reducing the amplitude of the ripple is that of simply increasing the discharge time constant. This can be achieved either by increasing the value of C_1 or by increasing the resistance value of R_L . In practice, however, the latter is not really an option because R_L is the effective resistance of the circuit being supplied and we don't usually have the ability to change it! Increasing the value of C_1 is a more practical alternative and very large capacitor values (often in excess of $4,700\ \mu\text{F}$) are typical.

. The amount of ripple is reduced by an approximate factor equal to

$$\frac{X_C}{\sqrt{R^2 + X_C^2}}$$



Example

The R – C smoothing filter in a 50 Hz mains operated half-wave rectifier circuit consists of $R1 = 100\ \Omega$ and $C2 = 1,000\ \mu\text{F}$. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.

Solution

$$\begin{aligned}X_C &= \frac{1}{2\pi fC} = \frac{1}{6.28 \times 50 \times 1,000 \times 10^{-6}} \\&= \frac{1,000}{314} = 3.18 \, \Omega\end{aligned}$$

The amount of ripple at the output of the circuit (i.e. appearing across C_1) will be given by:

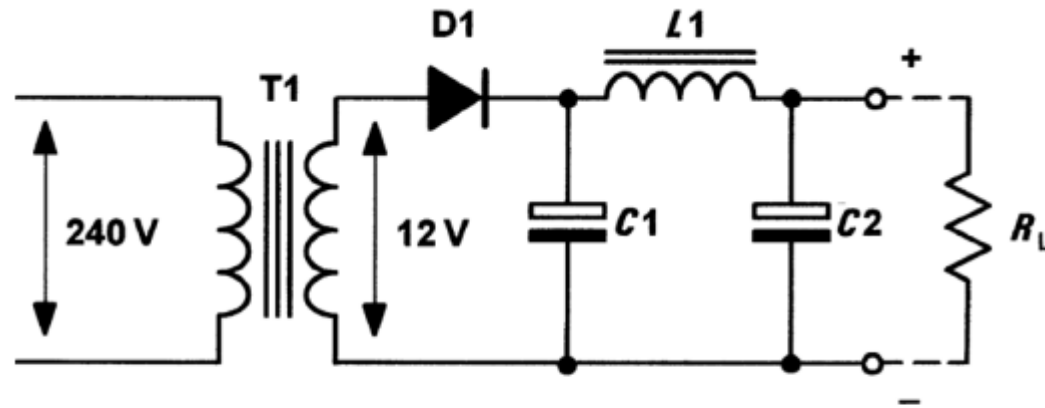
$$V_{\text{ripple}} = 1 \times \frac{X_C}{\sqrt{R^2 + X_C^2}} = 1 \times \frac{3.18}{\sqrt{100^2 + 3.18^2}}$$

From which:

$$V = 0.032 \, \text{V} = 32 \, \text{mV}$$

Improved ripple filters

A further improvement can be achieved by using an inductor, $L1$, instead of a resistor in the smoothing circuit. This circuit also offers the advantage that the minimum d.c. voltage is dropped across the inductor (the d.c. output voltage is reduced by an amount equal to the voltage drop across $R1$). Fig. below shows the circuit of a half-wave power supply with an L–C smoothing circuit. At the ripple frequency, $L1$ exhibits a high value of inductive reactance while $C1$ exhibits a low value of capacitive reactance. The combined effect is that of an attenuator which greatly reduces the amplitude of the ripple while having a negligible effect on the direct voltage.



Half-wave rectifier circuit with L–C smoothing filter

The L – C smoothing filter in a 50 Hz mains operated half-wave rectifier circuit consists of $L1 = 10$ H and $C2 = 1,000$ μ F. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.

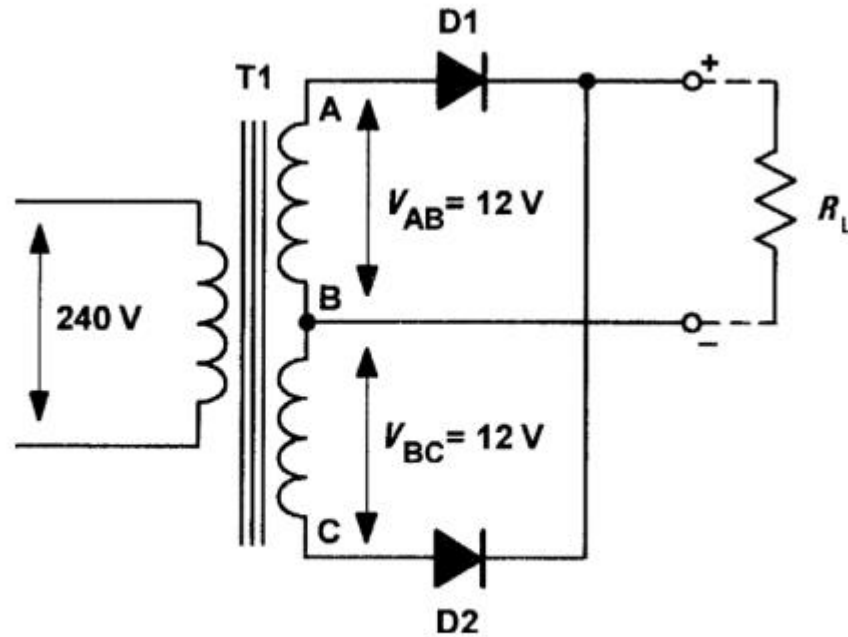
Solution

Once again, the reactance of the capacitor, $C1$, is 3.18 Ω (see Example 6.2). The reactance of $L1$ at 50 Hz can be calculated from:

$$X_L = 2\pi fL = 2 \times 3.14 \times 50 \times 10 = 3,140 \Omega$$

The amount of ripple at the output of the circuit (i.e. appearing across $C1$) will be approximately given by:

$$V = 1 \times \frac{X_C}{X_C + X_L} = 1 \times \frac{3.18}{3140 + 3.18} \approx 0.001 \text{ V}$$



Bi-phase rectifier circuit

Bi-phase rectifier circuits

Fig. shows a simple bi-phase rectifier circuit. Mains voltage (240 V) is applied to the primary of the step-down transformer (T1) which has two identical secondary windings, each providing 12 V r.m.s. (the turns ratio of T1 will thus be 240/12 or 20:1 for *each* secondary winding).

On positive half-cycles, point A will be positive with respect to point B. Similarly, point B will be positive with respect to point C. In this condition D1 will allow conduction (its anode will be positive with respect to its cathode) while D2 will not allow conduction (its anode will be negative with respect to its cathode). Thus D1 alone conducts on positive half-cycles.

On negative half-cycles, point C will be positive with respect to point B. Similarly, point B will be positive with respect to point A. In this condition D2 will allow conduction (its anode will be positive with respect to its cathode) while D1 will not allow conduction (its anode will be negative with respect to its cathode). Thus D2 alone conducts on negative half-cycles.

Fig. 6.11 shows the bi-phase rectifier circuit with the diodes replaced by switches. In Fig. 6.11(a) D1 is shown conducting on a positive half-cycle while in Fig. 6.11(b) D2 is shown conducting. The result is that current is routed through the load *in the same direction* on successive half-cycles.

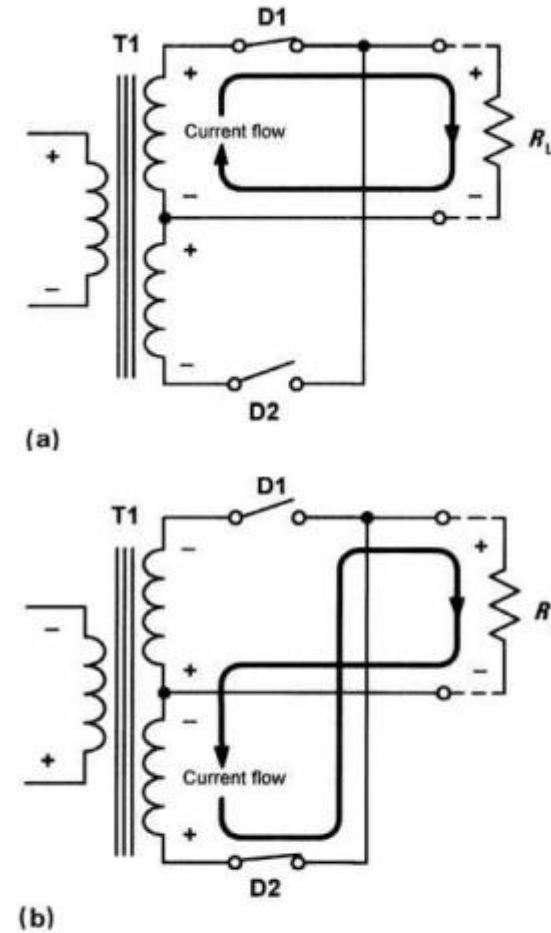
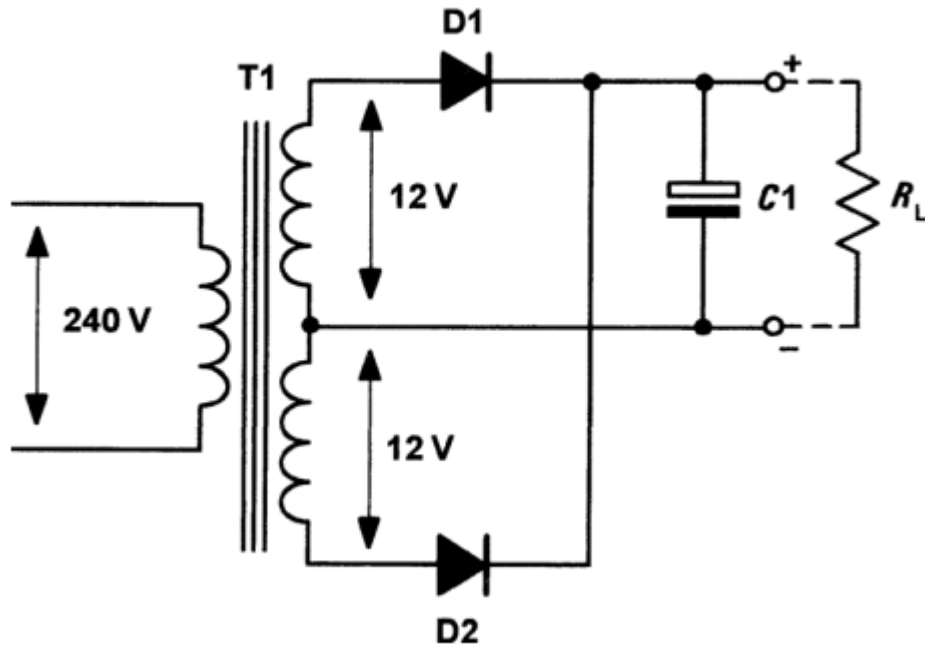
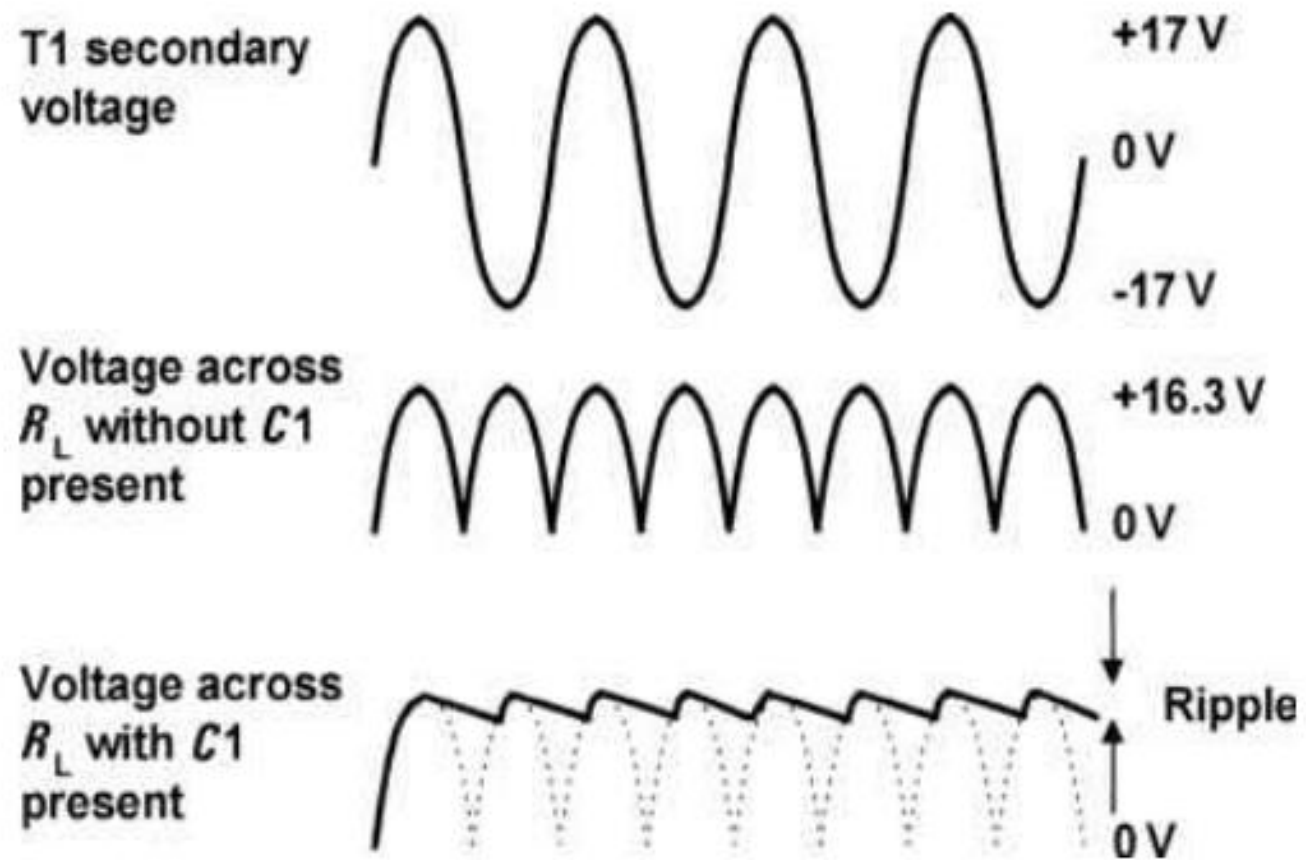


Figure 6.11 (a) Bi-phase rectifier with D1 conducting and D2 non-conducting
(b) bi-phase rectifier with D2 conducting and D1 non-conducting



Bi-phase rectifier with reservoir capacitor

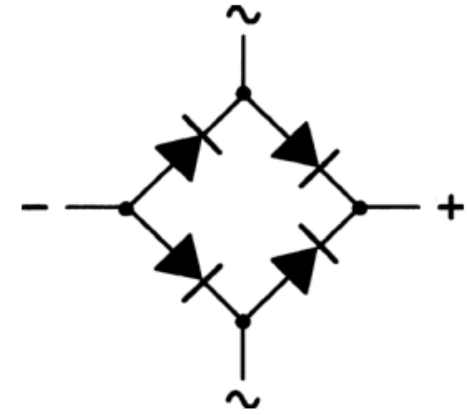
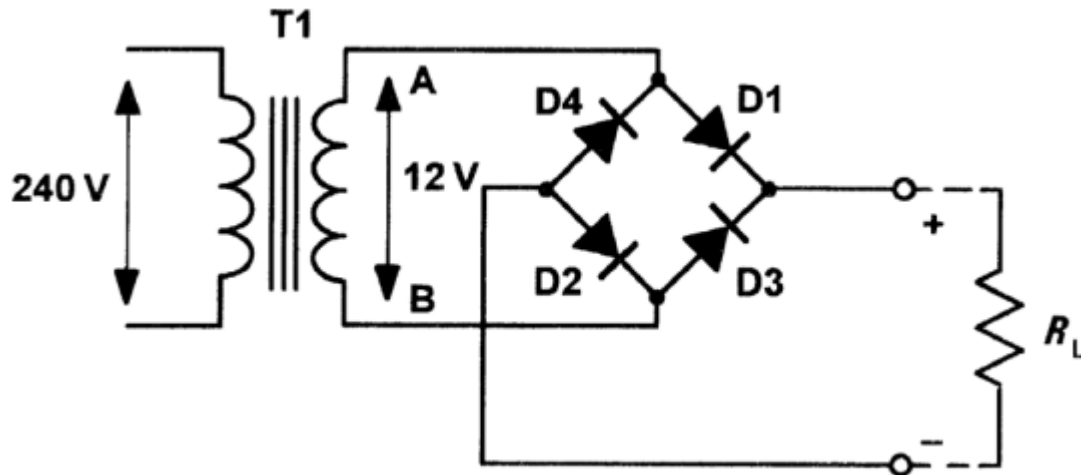
Figure shows how a reservoir capacitor (C1) can be added to ensure that the output voltage remains at, or near, the peak voltage even when the diodes are not conducting. This component operates in exactly the same way as for the half-wave circuit, i.e. it charges to approximately 16.3 V at the peak of the positive half-cycle and holds the voltage at this level when the diodes are in their non-conducting states. The time required for C1 to charge to the maximum (peak) level is determined by the charging circuit time constant (the series resistance multiplied by the capacitance value).



Waveforms for the bi-phase rectifier

Bridge rectifier circuits

An alternative to the use of the bi-phase circuit is that of using a four-diode bridge rectifier in which opposite pairs of diodes conduct on alternate half-cycles. This arrangement avoids the need to have two separate secondary windings. A full-wave bridge rectifier arrangement is shown in Figure. Mains voltage (240 V) is applied to the primary of a step-down transformer (T1). The secondary winding provides 12 V r.m.s. (approximately 17 V peak) and has a turns ratio of 20:1, as before. On positive half-cycles, point A will be positive with respect to point B. In this condition D1 and D2 will allow conduction while D3 and D4 will not allow conduction. Conversely, on negative half-cycles, point B will be positive with respect to point A. In this condition D3 and D4 will allow conduction while D1 and D2 will not allow conduction.



Four diodes connected as a bridge

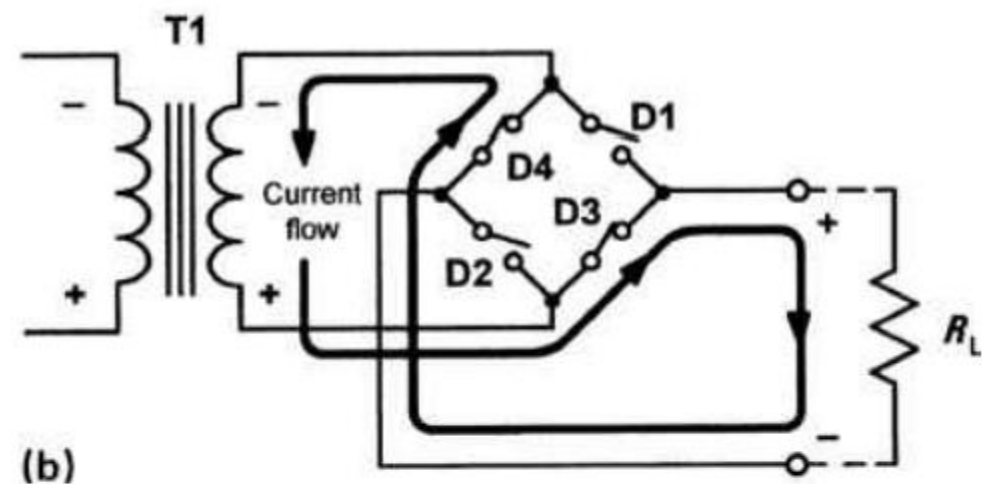
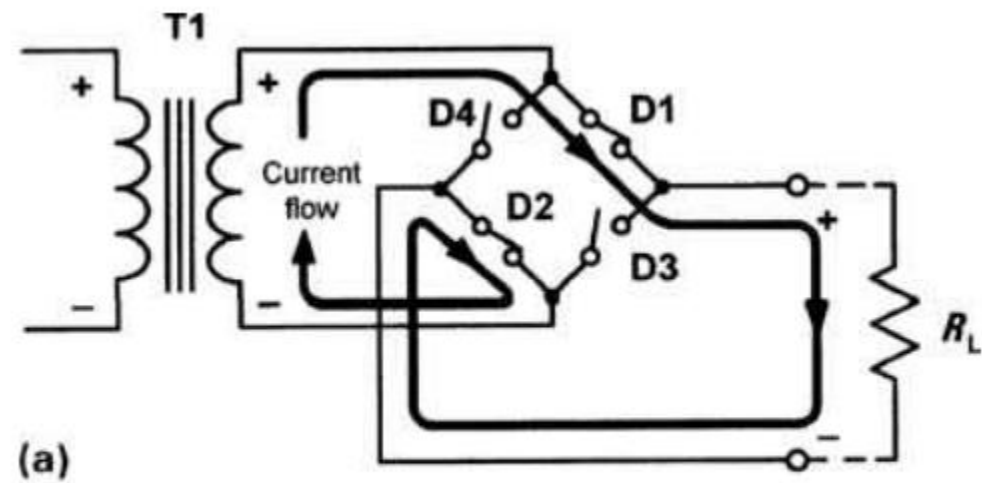


Figure 6.16 (a) Bridge rectifier with D1 and D2 conducting, D3 and D4 non-conducting (b) bridge rectifier with D1 and D2 non-conducting, D3 and D4 conducting

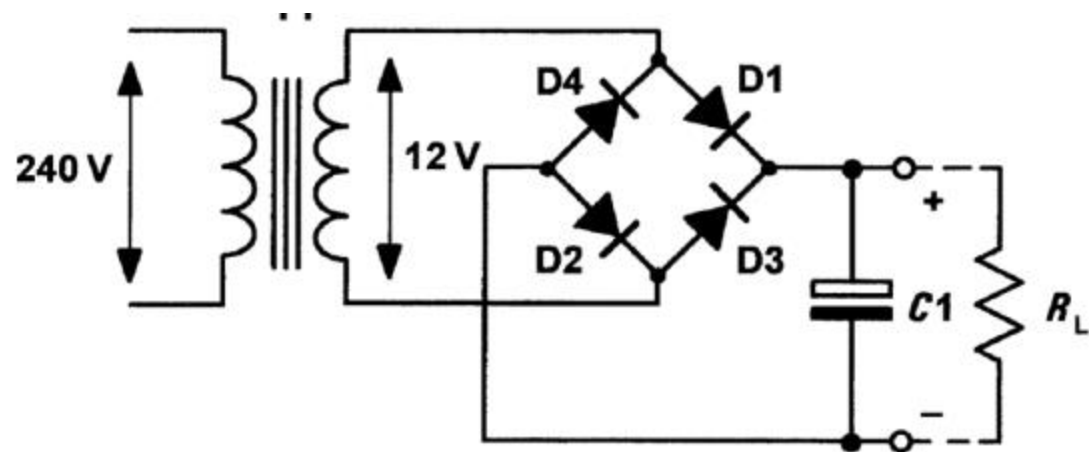


Figure 6.17 Bridge rectifier with reservoir capacitor

